WORKSHEET 1 SQL

1. Which of the following is/are DDL commands in SQL?

A) Create C) Delete D) ALTER

2. Which of the following is/are DML commands in SQL?

A) Update B) Delete C) Select

3. Full form of SQL is:

B) Structured Query Language

4. Full form of DDL is:

B) Data Definition Language

5. DML is:

A) Data Manipulation Language

6. Which of the following statements can be used to create a table with column B int type and C float type?

C) Create Table A (B int C float)

7. Which of the following statements can be used to add a column D (float type) to the table A created above?

B) Alter Table A ADD COLUMN D float

8. Which of the following statements can be used to drop the column added in the above question?

B) Alter Table A Drop Column D

9. Which of the following statements can be used to change the data type (from float to int) of the column D of table A created in above questions?

B) Alter Table A Alter Column D int

10. Suppose we want to make Column B of Table A as primary key of the table. By which of the following statements we can do it?

C) Alter Table A Add Primary key B

11. What is data-warehouse?

- A data warehouse (DW) is a digital storage system that connects and harmonizes large amounts of data from many different sources.
- Its purpose is to feed business intelligence (BI), reporting, and analytics, and support regulatory requirements – so companies can turn their data into insight and make smart, data-driven decisions.
- Data warehouses store current and historical data in one place and act as the single source of truth for an organization.
- A data warehouse provides the information for your data-driven decisions – and helps you make the right call on everything from new product development to inventory levels.

There are many benefits of a data warehouse. Here are just a few:

- **Better business analytics:** With data warehousing, decision-makers have access to data from multiple sources and no longer have to make decisions based on incomplete information.

- **Faster queries:** Data warehouses are built specifically for fast data retrieval and analysis. With a DW, you can very rapidly query large amounts of consolidated data with little to no support from IT.
- **Improved data quality:** Before being loaded into the DW, data cleansing cases are created by the system and entered in a worklist for further processing, ensuring data is transformed into a consistent format to support analytics – and decisions – based on high quality, accurate data.
- **Historical insight:** By storing rich historical data, a data warehouse lets decision-makers learn from past trends and challenges, make predictions, and drive continuous business improvement.

12. What is the difference between OLTP VS OLAP?

OLAP stands for On-Line Analytical Processing. It is used for analysis of database information from multiple database systems at one time such as sales analysis and forecasting, market research, budgeting and etc. Data Warehouse is the example of OLAP system.

OLTP stands for On-Line Transactional processing. It is used for maintaining the online transaction and record integrity in multiple access environments. OLTP is a system that manages very large number of short online transactions for example, ATM.

13. What are the various characteristics of data-warehouse?

various characteristics of data-warehouse are followings-

- Subject Oriented
    - A data warehouse is subject oriented because it provides information around a subject rather than the organization's ongoing operations.
    - These subjects can be product, customers, suppliers, sales, revenue, etc. A data warehouse does not focus on the ongoing operations, rather it focuses on modelling and analysis of data for decision making.
- Integrated
    - A data warehouse is constructed by integrating data from heterogeneous sources such as relational databases, flat files, etc.
    - This integration enhances the effective analysis of data.
- Time Variant
    - The data collected in a data warehouse is identified with a particular time period.
    - The data in a data warehouse provides information from the historical point of view.
- Non-volatile
    - Non-volatile means the previous data is not erased when new data is added to it.

14. What is Star-Schema??

A star schema is a database organizational structure optimized for use in a data warehouse or business intelligence that uses a single large fact table to store transactional or measured data, and one or more smaller dimensional tables that store attributes about the data.

- It is called a star schema because the fact table sits at the centre of the logical diagram, and the small dimensional tables branch off to form the points of the star.
- A fact table sits at the centre of a star schema database, and each star schema database only has a single fact table.
- The fact table contains the specific measurable (or quantifiable) primary data to be analysed, such as sales records, logged performance data or financial data.
- It may be transactional -- in that rows are added as events happen -- or it may be a snapshot of historical data up to a point in time.